

User Narrative Study for Dealing with Deceptive Chatbot Scams Aiming to Online Fraud

Omid Veisi
School of Information
University of Toronto
Toronto, Ontario, Canada
omid.veisi@mail.utoronto.ca

Khoshnaz Kazemian
Human Computer Interaction
University of Siegen
Siegen, Germany
khoshnaz.kazemian@student.uni-siegen.de

Farzaneh Gerami
Human Computer Interaction
University of Siegen
Siegen, NRW/Siegen, Germany
farzaneh.gerami@student.uni-siegen.de

Mahya Mirzaee Kharghani
Human Computer Interaction
University of Siegen
Siegen, Germany
mahya.mirzaeeKharghani@student.uni-siegen.de

Sima Amirkhani
University of Siegen
Siegen, Germany
sima.amirkhani@student.uni-siegen.de

Delong K. Du
Human Computer Interaction
University of Siegen
Siegen, North Rhine-Westphalia
Germany
delong.du@uni-siegen.de

Gunnar Stevens
Information Systems
University of Siegen
Siegen, Germany
gunnar.stevens@uni-siegen.de

Alexander Boden
Institute for Consumer Informatics
Bonn-Rhein-Sieg University of
Applied Science
Sankt Augustin, NRW, Germany
alexander.boden@h-brs.de

Abstract

Chatbots have become an integral part of everyday life, making it increasingly important to conduct empirical research on their use in the context of online scams. This study explores users' perceptions of deception by advanced chatbots based on semi-structured interviews with 16 individuals who believe they were victims of scam bots. The preliminary findings examine how chatbots can convincingly mimic human conversation, leading to deception. Results show how participants identified distinct differences in communication styles—such as mechanical responses, repetitive messaging, and emotional flatness—that shaped their perceptions and suspicions of interacting with automated chatbots rather than authentic humans.

CCS Concepts

• **Security and privacy** → **Social aspects of security and privacy**; Security requirements; • **Social and professional topics** → *Privacy policies*.

Keywords

Chatbots, Cybersecurity, Identity Fraud, Privacy, Online Social Media

ACM Reference Format:

Omid Veisi, Khoshnaz Kazemian, Farzaneh Gerami, Mahya Mirzaee Kharghani, Sima Amirkhani, Delong K. Du, Gunnar Stevens, and Alexander Boden. 2025. User Narrative Study for Dealing with Deceptive Chatbot Scams Aiming to Online Fraud. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3720152>

1 Introduction

While the rise of chatbots has brought numerous efficiencies to sectors like customer service and social media, their malicious applications are increasingly prevalent, e.g., the increasing use of bots for scamming purposes [9, 22, 24, 37, 49, 50], generating fake videos and images of misleading information [28], and generating fake news [7]. Growing concerns surrounding identity fraud through online social networks are escalating, highlighting the risks these automated programs pose to users [19]. Despite the utility of bots offer in enhancing operational efficiencies, malicious bots can be used in crimes online against vulnerable internet users [4]. Addressing such issues is crucial for understanding how malicious bots manipulate interactions and engage in deceptive practices. The study seeks to address a gap in understanding how users perceive and detect chatbot-driven deception by semi-structured interviewing 16 victims who believe they have experienced online chatbot scams. This qualitative narrative approach explores how frauds happen from the user's perspective, what they notice, and how we can assist in detecting them, guided by the following research question: **How do users differentiate between automated chatbots and genuine users?**

The research question aims to empirically understand current research on human-chatbot interactions and users' strategies they

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720152>

employ to ensure security. By exploring these aspects, the study seeks to contribute valuable insights to the field of cybersecurity from the perspective of human-computer interaction. While prior work covers chatbot design and malicious use cases [1, 39, 61], the human perspective on chatbot deception remains underexplored. This study bridges this gap by examining how users perceive and detect scams aglining with traditional *Turing's Test* on Artificial Intelligence, contributing to both cybersecurity and human-computer interaction (HCI) research. It provides empirical insights into how users perceive and identify chatbot scams, as outlined below:

- Offering insights into how users perceive and identify chatbot scams.
- Highlighting communication patterns that trigger suspicion (e.g., repetitive messaging, emotional flatness).
- Providing practical recommendations for improving scam detection of chatbot.

2 Related Work

2.1 Chatbot Scams and Cybersecurity Risks

Nowadays, chat conversations reign supreme with the rise of the internet and social media. As a result, many technologists envisioned chatbots as a way for humans to communicate with computers. Many companies and services use bots and chatbots to respond to the user efficiently and productively [12], such as bank services [23, 25], health care [26], and libraries [36]. Chatbot design principles (transparency, responsiveness, and humanlike anthropomorphism) must balance functionality with security to mitigate malicious exploitation. Malicious chatbots leverage AI/NLP to mimic humans, enabling scams, phishing, identity theft, and social engineering. These systems pose risks such as privacy breaches, financial losses, and botnet-driven attacks (e.g., DDoS, spam extraction) [1, 39, 61]. LLM-driven chatbots are increasingly convincing, complicating fraud detection [45]. On top of that, social bots infiltrate online social networks (OSNs) to harvest personal data (emails, phone numbers) for spam, phishing, or political manipulation (e.g., astroturfing, smear campaigns) [42, 62]. Over 270 million Facebook accounts are fraudulent social media accounts that may involve bots [51]. Research shows an online tool that follows political memes on Twitter and detects astroturfing, smear campaigns, and other disinformation in U.S. elections [42]. Chatbots in "cybernetic space" disrupt offline/online environments, enabling identity fraud, spam, and system attacks [2, 61]. Neural generative agents excel in informal conversations but require further research to assess real-world efficacy in adversarial contexts (e.g., interactions with motivated attackers) [48]. The analysis of social network messages is critical for threat prevention. Future work must prioritize detecting malicious chatbot use, improving trauma-informed design to prevent harm beyond financial loss [6], and balancing humanlike quality with security objectives [40].

2.2 User Psychological Vulnerability

Online scam victims' suspicion levels affect their ability to distinguish human vs. chatbot perpetrators; however, financial or emotional motives (e.g., urgency, greed) often impair judgment [11, 52], while fraudulent activities harm psychological/physical health and erode trust in online platforms [53]. Trauma-informed computing

addresses these risks by adapting psychological safety principles to UX design, security, and AI, fostering safer digital environments perceived as more usable than traditional resources [15, 58]. Specifically in Human-computer interaction (HCI) research, existing bodies of research highlight how empathetic and anthropomorphic conversational agents (CAs) like Shing exploit psychological engagement to influence behavior. For instance, Shing's empathetic design encourages users to terminate risky transactions by leveraging trust and emotional connection, achieving higher engagement than rule-based systems [21]. However, this human-like interaction risks exploiting emotional vulnerability, particularly in identity fraud and consumer scams [4, 17, 35]. Emotion-aware conversational models must prioritize understanding emotion causes to build genuine empathy [27]. Over-reliance on chatbots for repetitive tasks risks dehumanizing communication [33], underscoring the need for empathetic AI that mitigates exploitation while enhancing engagement.

3 Method

3.1 Participants: Online Chatbot Scam Victims

Shown in Table 1, we recruited 16 participants who were recruited for their past experiences with fraud through social media platforms, such as Telegram, Instagram, Twitter, WhatsApp, LinkedIn, and university groups. Participants were aged 24-35 years ($M=29.6$). They represented diverse professional backgrounds, including technology and IT roles (6), design and engineering (3), business and finance (3), education (2), and other fields (2). Participants reported experiences with impersonating chatbots across different platforms, primarily Telegram (7), Instagram (3), websites (3), and mixed platforms, including Facebook and Gmail (3). The recency of these experiences varied, with 5 participants encountered such chatbots within the previous 3 months, 4 participants between 3-12 months prior, 4 participants between 1-3 years ago, and 3 participants more than 4 years ago. This diversity in encounter timing allowed us to explore both recent and more historical perspectives on chatbot impersonation experiences. Semi-structured interviews are details regarding participant demographics and the types of fraud experienced. Participants reported encountering various fraud types, with investment fraud being the most common, affecting four individuals primarily on Telegram. Other prevalent categories included relationship and trust fraud (romance fraud) and identity fraud, often employing emotionally manipulative profiles tailored to the target's preferences [10, 17]. P4, P9, and P14 were interviewed in person and recorded using an iPhone 14 Pro Max, while the rest were interviewed online via Zoom. Each interview lasted between 40 minutes and 1 hour. Since the study topic involves sensitive narratives, ethical concerns need to be taken seriously [5]. The study obtained ethical approval from the university's ethical board, and participants provided their consent by signing consent forms. Throughout the semi-structured interview process, we maintained strict ethical protocols, ensuring participant privacy and offering options to pause or stop interviews at any time.

3.2 Data Collection: Interview Procedure

We started the interviews with a short explanation of the research objectives and guidelines, followed by demographic questions. After

Table 1: Participants of Semi-Structured Interviews

ID	Age	Sex	Occupations	Platform	Whom Chatbot Impersonates	Time
P1	28	M	Front-End Developer	Telegram	Acquaintance	3 years ago
P2	28	M	Architect	Website	Government Officer	1 year ago
P3	30	F	Architect	Telegram	Finance Trader	3 months ago
P4	30	M	Master Student	Instagram	Finance Trader	2 months ago
P5	27	M	IT Engineering	Telegram	Finance Trader	3 months ago
P6	33	M	Water Engineering	Telegram	Romance Interests	6 months ago
P7	32	M	Network Security	Telegram	Bank Staff	9 months ago
P8	35	M	ERP Specialist	Websites	Betting Service	1 year ago
P9	27	M	Research Associate	Websites	Acquaintance	3 years ago
P10	33	F	Teacher	Instagram	Famous Celebrity	1 week ago
P11	33	M	Master Student	Gmail & Social Media	Wealthy Senior Seeking Friends Online	10 years ago
P12	28	F	Bank Employee	Facebook & Telegram	Job Recruiter	3 months ago
P13	32	F	Teacher	Facebook	Job Recruiter	8 months ago
P14	24	F	Master Student	Telegram	Job Recruiter	3 years ago
P15	28	M	Business Coach	Instagram & Telegram	Famous Celebrity	2 years ago
P16	26	M	Sales Manager	Instagram	Wealthy Senior Seeking Friends Online	4 years ago

that, the interview script addressed three main topics: (i) a discussion about fraud; (ii) the main challenges, including challenges before, during, and after the crime; and (iii) the solutions to those challenges. During the initial segment of the interview, we inquired about the narrative surrounding the occurrence of online fraud and endeavored to comprehend the user's response to it. In this regard, we presented a series of questions that aimed to elicit a detailed account of the experience and the user's emotional and behavioral reactions towards the fraud. The list of these semi-structured interview questions can be found below:

- When did this fraud happen? How did this fraud happen? Please tell us as detailed as you can remember. taking us through the events and how you felt.
- Which chatbot platform or application were you using when the incident took place?

During the second step, our focus was on identifying the primary challenges associated with the crime, both before, during, and after its occurrence. To gain a deeper understanding of how individuals can discern the presence of a chatbot and the associated emotions, they may experience the following questions:

- How do you feel about the overall story and your experience with identity theft?
- How do you now identify and recognize potential scammers in light of this incident?
- Have you made any changes to your habits or behavior as a result of this incident?

Lastly, we seek advice from the interviewee to share prevention strategies based on their firsthand encounter with identity theft with the following questions:

- Do you believe that special skills are required to recognize scammers in general? Why?
- What advice would you offer to others based on your experience with identity theft?
- If given the opportunity, what changes or improvements would you like to see made to the bot?

With those interview questions, we prompted them to revisit their experiences as a narrative series of events [8]. Our interview protocol was designed to elicit detailed accounts of the arc of their encounters with impersonating chatbots, from initial contact through the realization of deception to aftermath and reflection.

3.3 Data Analysis: Understanding Online Chabot Scam Experience

For data analysis, we conducted a narrative thematic analysis approach to deeply understand participants' experiences with chatbot impersonation [8]. Rather than merely categorizing experiences, we sought to capture the chronological unfolding of events while preserving the richness of participants' subjective experiences and meaning-making processes. Following data collection, our analysis proceeded through several phases. Researchers individually conducted an initial narrative analysis for four participants, constructed individual storylines, and identified key narrative elements. The team then engaged in co-analysis collaborative sessions to share findings, reconcile interpretations, and identify overarching narrative patterns across accounts.

Integrating both deductive and inductive analytical strategies, this narrative thematic analysis approach revealed temporal patterns that conventional thematic analysis might overlook. It highlighted how fraudulent encounters unfold as dynamic processes rather than static experiences, with important implications for developing preventative and remedial strategies that address the sequential nature of online deception. This approach allowed us to construct a nuanced understanding of chatbot impersonation as a lived experience, revealing not just what happened to participants but how they made sense of these events within their broader life narratives.

4 Results

4.1 Deceptive Chatbots Impersonate Different Identity

The findings reveal how chatbots attempt to exploit the psychological vulnerabilities of users, showcasing how technology can

manipulate users during critical activities and highlighting the adaptability and versatility of chatbots in exploiting human behavior across various contexts. For example, chatbots often pose as job recruiters, deceiving users into applying for fake positions and exploiting their job-seeking efforts. Participants reported receiving identical messages from automated bots after submitting applications, which exposed their personal information. Additionally, chatbots frequently take on the roles of finance traders, misleading users with promises of profitable trading signals. Participants reported receiving enticing offers that capitalized on their desire for quick financial gains. Furthermore, chatbots impersonate famous celebrities or acquaintances, crafting familiar usernames and profiles to enhance their authenticity.

4.1.1 Chatbot Impersonates Job Recruiters. Several participants noted that chatbots can exploit users' job-seeking actions on websites (P12, P13, and P14) which are supposed to help users in finding a job. P14 mentioned: *"My friend and I were searching for a job, and we came across the DIVAR website where job offers are posted. As we had a good command of English and the job ad we saw was related to translating articles, we decided to apply for that job offer.[...] My friend and I applied for a job and received identical messages from the recipient ID. We later discovered that it was an automated bot."* Also, P12 reported the incident of faking hiring as she shared detailed personal information for employment, while P13 and P14 were the victims of con artists who claimed to offer jobs in exchange for money but ultimately left them with nothing and were blocked.

4.1.2 Chatbot Impersonates Wealthy Seniors for Online Friendship. Some users, P16 and P11, experienced the exploitation of emotional triggers, where chatbots created elaborate backstories of personal tragedy and wealth, preying on users' empathy and compassion. This way, chatbots try to establish an emotional connection and foster trust, thus aiming at enhancing the effectiveness of the fraud scheme. For example, P16 mentioned: *"The messages were that this lady had lost her husband. She said that she is suffering from cancer and will probably die in the next few months. She has a lot of wealth and wants to dedicate this wealth to the needy."* This highlights the nature of chatbot manipulation, in which scripted narratives combining personal tragedy, financial prospects, and companionship manipulate users' emotions and cultivate a sense of trust.

4.1.3 Chatbot Impersonates Finance Traders. P3 and P5 shared experiences with deceptive chatbots who offered profitable trading opportunities, capitalizing on users' avarice and the promise of quick monetary gains. The intricate designs of these chatbots allow for a seamless transition between automated responses and human-like interactions, luring users into fraud. P3 noted *"At 9:00 PM on the Telegram group, three different chatbots with different names and profile pictures send a pre-written message about purchasing a Bitcoin that they have built themselves. They never reply to other users. I realized these are chatbots. When I sent a message in private, and the chatbot forwarded again the same message, immediately."* This strategy frequently conceals the bot's original purpose from the user. P5 and P16 serve as examples of how chatbots are skilled at manipulating human psychology through the use of financial incentives. P5 mentioned *"They provided us with buy and sell signals for Bitcoin and other currencies listed on their exchange. The signals*

generated high profits, perhaps around 14% in just 30 minutes. This sparked a sense of greed."

4.1.4 Chatbot Impersonates Famous Celebrities or Acquaintances. Both P1 and P10 pointed out that *"chatbots mimic usernames and profiles to resemble human users, creating an illusion of authenticity that fosters familiarity and trust among potential victims"*. This strategy aligns with P15's experiences, which show that chatbots strategically select usernames resembling reputable entities to evoke a sense of credibility. P15 mentioned: *"This chatbot chose its name similar to famous and reputable exchanges in the world, which made users who search for its name think that it is the same reputable exchange, and many times unconsciously, the user trusts this chatbot after seeing this famous name."*

4.2 Users Realise Deceptive Chatbots

With regard to how users identify chatbots, participants noted that chatbots sometimes demonstrate predictable patterns in their messaging, such as consistent timing and immediate responses, which reveal their non-human nature and inability to effectively handle off-topic inquiries, often leading to formulaic replies and synchronized online behavior. Participants highlighted the noticeable differences between human and machine responses, pointing out that chatbots exhibit mechanical communication patterns, repetitive messaging, and a lack of emotional depth, which ultimately raises doubt and skepticism. Additionally, when users probe the bots with off-topic questions, it often reveals the chatbots' inability to respond appropriately, solidifying users' understanding that they are engaging with a machine rather than a real person.

4.2.1 Real human cannot immediately respond unless it's a chatbot. P1 emphasizes the chatbot's consistent message at specified times—a behavior that is discordant in true human encounters. *"It's amazing how she would send me a message at the same time every night."* Participant P15 shared an observation about the immediacy of chatbot responses: *"I think it was a system whose job was to send me a message as soon as I sent a message."* This highlights how bots often respond with unusual promptness, a key indicator of their non-human nature. Furthermore, P15 noted the strategy of probing bots with off-topic queries: *"I try to put pressure on the chatbot with a few messages and ask questions outside of the area it was programmed for [...] you can tell if it is a chatbot or not by the way it answers."* This method reveals the inability of bots to adequately handle unexpected or unrelated topics, often leading to formulaic or irrelevant responses, a clear temporal pattern distinguishing them from human users. Participant P6 added another dimension to temporal patterns: *"Every time I was online, she was also online [...] it was like someone else was controlling this account."* This observation suggests that bot-controlled accounts may exhibit synchronized online patterns with the user, unlike the more random presence patterns typical of human users.

4.2.2 Humans don't constantly repeat themselves unless it's a chatbot. P9 and P10 emphasize the clear distinction between human and machine responses. P9 elucidates chatbots' original programming to mimic human intellect and behavior, and, as P10 emphasizes, this purpose is visible in the starkly mechanical aspect of their communication. Similarly, when inquiries are made in multiple languages,

P4 reveals discordant responses in chatbot conversations, casting doubt on the nature of the interaction. “[...] However, when I was asking for details, I kept getting the same message repeatedly. It was only later that I realized that I was talking to a chatbot and that I had been scammed.” By combining the experiences of the interviewees, a discernible pattern emerges, revealing warning indicators that elicit doubt and skepticism. Interviewees P6 and P11 found a pattern of limited and repetitive communication strategies used by chatbots. P6 noticed suspicions due to single-sentence responses and monotonous repetition, while P11 noted a lack of human emotion in messages. “Another thing that made me dubious was that she used to write single sentences and didn’t completely converse.” P13 also observed brief and formal responses, a common behavior in chatbot encounters: “Throughout our chat, she continued responding briefly, like a computer or a robot.”

5 Discussion

The differentiation between automated chatbots and genuine users relies on various communication patterns and behavioral cues. The study participants identified specific indicators that reveal the mechanical and predictable nature of chatbots, helping them discern their interactions. One of the key factors in the detection of deceptive chatbots is using a discernible pattern [41, 46, 64]. Chatbots use single-sentence responses and monotonous repetition to create a connection during chatting. This observation aligns with existing research on chatbot detection, emphasizing the significance of identifying formulaic responses as a primary indicator of automated activity [14, 16, 59]. Our findings indicate that suspicious behavior, such as single-sentence responses, repetitive communication, or inconsistencies in responses, should trigger red flags. Weiss noted that the content of many of the fraudulent comments is easily distinguished by the technique known as synonym replacement, which involves using search and replace to generate variations of a single sentence, thereby confirming our findings [59]. Participants highlighted the mechanical nature of chatbot responses, which tend to lack the emotional depth and variation expected in human communication. Moreover, the recognition of a noticeable temporal pattern in chatbot interactions reveals the constraints of automated systems in managing spontaneous or unexpected discussion subjects in online fraud, which previous researchers have proved [30, 34, 60]. Alternatively, the ability of chatbots to generate identical messages to multiple users at a specific time reveals a systematic approach that distinguishes them from genuine communicators [18]. The participants in the study noted the chatbot’s prompt reply to messages, highlighting the chatbot’s limited ability to handle unexpected or unrelated topics. As a result, chatbots often provide formulaic or irrelevant responses that can be easily distinguished from those of human users due to a clear temporal pattern [38, 43, 44].

Furthermore, when users posed off-topic questions, chatbots frequently struggled to provide appropriate responses, further affirming users’ suspicions about their non-human identity. The difference in language between machine and human responses plays a vital role in identifying if an interaction is genuine or artificial [63]. Likewise, the observations made by participants emphasize the significance of linguistic analysis in detecting bots, as inconsistencies in conversational style and coherence can reveal signs

of automated activity. Moreover, many participants noted that the quick response times of chatbots, such as immediate replies at the same hour each day, created an unmistakable distinction from human behavior. Long’s research into the chatbots on social media and the accessibility of chatbot creation tools emphasizes the need for users to be attentive to unusual behavioral patterns [29]. Also, participants (P6, P13, P10, and P14) pointed out certain suspicious behaviors, such as instant replies, that don’t fit with normal human interactions. Users should be careful when chatbots give inappropriate responses, like repeating themselves, sending messages that don’t make sense, ignoring questions, or giving too many compliments [32]. Researchers suggest finding chatbots based on text repetition [20]. This shows how important it is for users to notice chatbots that use the same phrases or statements over and over, which is a sign of automated responses.

Several researchers mentioned the importance of temporal patterns in chatbot activities. One of these studies was conducted by Chavoshi et al. They discovered repeating behavior, and anomalous behavior in the activities of Twitter bots and explained the significance of these temporal patterns in gaining a competitive advantage over humans [13]. Current research is working to address the challenging issue of chatbot mimicry by analyzing individual user patterns. In their research, Stine et al. [55] and Stine and Agarwal [54] delved into the information foraging behavior of bots by examining changes in vocabulary usage over time. They mentioned that chatbots write about a variety of topics each day, resulting in high surprises in their word usage patterns. In a study conducted by Al-Khateeb and Agarwal, it was discovered that Chatbot used seemingly legitimate usernames on Twitter to coordinate between groups [3]. Our study also confirms this and shows that participants who experienced identity fraud by chatbots underscore the significance of verifying the identity of online contacts. They noted that cross-referencing information and looking for unusual profile details can help achieve this. The findings suggest that while chatbots may adopt human-like profiles to create an illusion of authenticity, their limitations become evident during conversations, ultimately leading users to recognize them as bots. Drawing from the insights gained through previous encounters, the continued practice of skepticism and vigilance plays a pivotal role in recognizing and mitigating potential instances of chatbot-driven fraud within the digital landscape. Collectively, these guidelines furnish a proactive framework for navigating the hazards posed by deceptive chatbots, thereby enhancing the safety of online experiences [19][31][56].

6 Limitation and Future Works

Our narrative study just focuses on the victim and what happened during the fraud. However, future research can focus on the expert and hacker (scammers) interviews before, during, and after fraud. Furthermore, this research was conducted when tech companies such as OpenAI introduced ChatGPT O1, and Google introduced Gemini 2. However, we were unable to locate any examples of LLM programs being used to target individuals, possibly because this type of chatbot is relatively new. Future works can focus on the possibility of using LLM for scams and solutions to that. Therefore, future research endeavors can investigate various aspects of

chatbot and chatbot interactions, ethics, and their impact on society based on the narratives of experts. Additionally, future studies should explore how factors such as gender, age, socioeconomic status, and cultural background influence susceptibility to chatbot scams, as well as how these demographics are targeted by malicious actors using advanced AI technologies. To analyze these narratives, methods such as narrative coding [47] can be employed to systematically identify themes, patterns, and insights from qualitative data, providing a deeper understanding of the psychological and social dynamics at play in chatbot-related fraud.

7 Conclusion

This study investigates how users differentiate between automated chatbots and genuine users within the context of online scams, utilizing qualitative interviews with 16 victims of scammer bots. The results reveal that users depend on specific communication patterns and behavioral cues to identify chatbots, such as predictable messaging and immediate responses that are atypical of human interactions. Participants highlighted the mechanical nature of chatbot responses, which often lack the emotional depth and variation that characterize genuine conversations, leading users to recognize their non-human identity. The ability of chatbots to generate identical messages and exhibit quick response times further distinguished them from real users, while repetitive messaging contrasted sharply with the more dynamic conversational styles expected from humans. Ultimately, our findings suggest that while chatbots may attempt to create an illusion of authenticity, their limitations become evident during interactions, prompting users to remain vigilant and question the authenticity of their engagements with these automated entities.

Acknowledgments

This work was supported by the German Federal Ministry of Education and Research (BMBF) as part of the AntiScam project [57] (Grant NO. 16KIS2214).

References

- [1] Nitin Agarwal, Samer Al-Khateeb, Rick Galeano, and Rebecca Goolsby. 2017. Examining the use of botnets and their evolution in propaganda dissemination. *Defence Strategic Communications* 2, 1 (2017), 87–112. doi:10.30966/2018.riga.2.4
- [2] Samer Al-Khateeb and Nitin Agarwal. 2015. Examining botnet behaviors for propaganda dissemination: A case study of isil's beheading videos-based propaganda. In *2015 IEEE International conference on data mining workshop (icdmw)*. IEEE, 51–57. doi:10.1109/ICDMW.2015.41
- [3] Samer Al-Khateeb and Nitin Agarwal. 2016. UNDERSTANDING STRATEGIC INFORMATION MANOEUVRES IN NETWORK MEDIA TO ADVANCE CYBER OPERATIONS: A CASE STUDY ANALYSING PRO-RUSSIAN SEPARATISTS' CYBER INFORMATION OPERATIONS IN CRIMEAN WATER CRISIS. *Journal on Baltic Security* 2, 1 (2016).
- [4] Fatemeh Alizadeh, Gunnar Stevens, Timo Jakobi, and Jana Krüger. 2023. Catch Me If You Can: "Delaying" as a Social Engineering Technique in the Post-Attack Phase. *Proc. ACM Hum.-Comput. Interact.* 7, CSCW1, Article 32 (apr 2023), 25 pages. doi:10.1145/3579465
- [5] Sima Amirkhani. 2023. Taking ethics seriously. In *Torn Many Ways: Politics, Conflict and Emotion in Research*. Springer, 105–119.
- [6] Sima Amirkhani, Fatemeh Alizadeh, Dave Randall, and Gunnar Stevens. 2024. Beyond Dollars: Unveiling the Deeper Layers of Online Romance Scams Introducing "Body Scam". In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–6.
- [7] Dipto Barman, Ziyi Guo, and Owen Conlan. 2024. The dark side of language models: Exploring the potential of llms in multimedia disinformation generation and dissemination. *Machine Learning with Applications* (2024), 100545.
- [8] Tea Torbenfeldt Bengtsson and Ditte Andersen. 2020. Narrative analysis: Thematic, structural and performative. *Qualitative Analysis: Eight Approaches for the Social Sciences: Järvinen, M., Mik-Meyer, N., Eds (2020)*, 265–282.
- [9] Ivo Benke, Michael Thomas Knierim, and Alexander Maedche. 2020. Chatbot-Based Emotion Management for Distributed Teams: A Participatory Design Study. *Proc. ACM Hum.-Comput. Interact.* 4, CSCW2, Article 118 (oct 2020), 30 pages. doi:10.1145/3415189
- [10] Alexander Bilz, Lynsay A Shepherd, and Graham I Johnson. 2023. Tainted Love: A Systematic Review of Online Romance Fraud. *arXiv preprint arXiv:2303.00070* (2023). doi:10.48550/arXiv.2303.00070
- [11] Mark Button and Cassandra Cross. 2017. *Cyber frauds, scams and their victims*. Taylor & Francis. doi:10.4324/9781315679877
- [12] Ana Paula Chaves, Jesse Egbert, Toby Hocking, Eck Doerry, and Marco Aurelio Gerosa. 2022. Chatbots Language Design: The Influence of Language Variation on User Experience with Tourist Assistant Chatbots. *ACM Trans. Comput.-Hum. Interact.* 29, 2, Article 13 (jan 2022), 38 pages. doi:10.1145/3487193
- [13] Nikan Chavoshi, Hossein Hamooni, and Abdullah Mueen. 2017. Temporal Patterns in Bot Activities. In *Proceedings of the 26th International Conference on World Wide Web Companion* (Perth, Australia) (WWW '17 Companion). International World Wide Web Conferences Steering Committee, Republic and Canton of Geneva, CHE, 1601–1606. doi:10.1145/3041021.3051114
- [14] Bocheng Chen, Guangjing Wang, Hanqing Guo, Yuanda Wang, and Qiben Yan. 2023. Understanding multi-turn toxic behaviors in open-domain chatbots. In *Proceedings of the 26th International Symposium on Research in Attacks, Intrusions and Defenses*. 282–296.
- [15] Janet X Chen, Allison McDonald, Yixin Zou, Emily Tseng, Kevin A Roundy, Acar Tamersoy, Florian Schaub, Thomas Ristenpart, and Nicola Dell. 2022. Trauma-informed computing: Towards safer technology experiences for all. In *Proceedings of the 2022 CHI conference on human factors in computing systems*. 1–20.
- [16] Wentao Chen, Fuzhou Wang, and Matthew Edwards. 2023. Active Countermeasures for Email Fraud. In *2023 IEEE 8th European Symposium on Security and Privacy (EuroSecP)*. IEEE, 39–55. doi:10.1109/EuroSP57164.2023.00012
- [17] Joel Cherus, Jason Githeko, Joseph Siror, and Kageni Njagi. 2014. Identity fraud: A literature review and future research directions. (2014). <http://repository.must.ac.ke/handle/123456789/229>
- [18] Ilias Dimitriadis, Marinos Poititis, Christos Faloutsos, and Athena Vakali. 2022. TG-OUT: temporal outlier patterns detection in Twitter attribute induced graphs. *World Wide Web* 25, 6 (2022), 2429–2453.
- [19] Eric Ferreira Dos Santos, Danilo Carvalho, Livia Ruback, and Jonice Oliveira. 2019. A New Approach to Bot Detection A Transparency-Focused Approach. In *Companion Proceedings of The 2019 World Wide Web Conference* (San Francisco, USA) (WWW '19). Association for Computing Machinery, New York, NY, USA, 545–552. doi:10.1145/3308560.3317599
- [20] Mehdi Golzadeh, Alexandre Decan, and Natarajan Chidambaram. 2022. On the accuracy of bot detection techniques. In *Proceedings of the Fourth International Workshop on Bots in Software Engineering*. 1–5. doi:10.1145/3528228.3528406
- [21] Jingya Guo, Jiajing Guo, Changyuan Yang, Yanjing Wu, and Lingyun Sun. 2021. Shing: A Conversational Agent to Alert Customers of Suspected Online-payment Fraud with Empathetical Communication Skills. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 383, 11 pages. doi:10.1145/3411764.3445129
- [22] Soomin Kim, Jinsu Eun, Changhoon Oh, Bongwon Suh, and Joonhwan Lee. 2020. Bot in the Bunch: Facilitating Group Chat Discussion by Improving Efficiency and Participation with a Chatbot. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (<conf-loc>, <city>Honolulu</city>, <state>HI</state>, <country>USA</country>, </conf-loc>) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3313831.3376785
- [23] Songhyun Kim, Junseok Goh, and Soojin Jun. 2018. The Use of Voice Input to Induce Human Communication with Banking Chatbots. In *Companion of the 2018 ACM/IEEE International Conference on Human-Robot Interaction* (Chicago, IL, USA) (HRI '18). Association for Computing Machinery, New York, NY, USA, 151–152. doi:10.1145/3173386.3176970
- [24] Lorenz Cuno Klopfenstein, Saverio Delpriori, Silvia Malatini, and Alessandro Bogliolo. 2017. The Rise of Bots: A Survey of Conversational Interfaces, Patterns, and Paradigms. In *Proceedings of the 2017 Conference on Designing Interactive Systems* (Edinburgh, United Kingdom) (DIS '17). Association for Computing Machinery, New York, NY, USA, 555–565. doi:10.1145/3064663.3064672
- [25] Chaitrali S Kulkarni, Amruta U Bhavsar, Savita R Pingale, and Satish S Kumbhar. 2017. BANK CHAT BOT—an intelligent assistant system using NLP and machine learning. *International Research Journal of Engineering and Technology* 4, 5 (2017), 2374–2377.
- [26] Minha Lee, Sander Ackermans, Nena van As, Hanwen Chang, Enzo Lucas, and Wijnand IJsselstein. 2019. Caring for Vincent: A Chatbot for Self-Compassion. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300932

- [27] Yanran Li, Ke Li, Hongke Ning, Xiaoqiang Xia, Yalong Guo, Chen Wei, Jianwei Cui, and Bin Wang. 2021. Towards an online empathetic chatbot with emotion causes. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2041–2045. doi:10.1145/3404835.3463042
- [28] Yixuan Li, Xuelin Liu, Xiaoyang Wang, Shiqi Wang, and Weisi Lin. 2024. FakeBench: Uncover the Achilles' Heels of Fake Images with Large Multimodal Models. *arXiv preprint arXiv:2404.13306* (2024).
- [29] Kiel Long, John Vines, Selina Sutton, Phillip Brooker, Tom Feltwell, Ben Kirman, Julie Barnett, and Shaun Lawson. 2017. "Could You Define That in Bot Terms"? Requesting, Creating and Using Bots on Reddit. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 3488–3500. doi:10.1145/3025453.3025830
- [30] Michele Mazza, Stefano Cresci, Marco Avvenuti, Walter Quattrociocchi, and Maurizio Tesconi. 2019. Rtbust: Exploiting temporal patterns for botnet detection on twitter. In *Proceedings of the 10th ACM conference on web science*. 183–192.
- [31] Fred Morstatter, Liang Wu, Tahora H Nazer, Kathleen M Carley, and Huan Liu. 2016. A new approach to bot detection: striking the balance between precision and recall. In *2016 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*. IEEE, 533–540. doi:10.1109/ASONAM.2016.7752287
- [32] Andreea Muresan and Henning Pohl. 2019. Chats with Bots: Balancing Imitation and Engagement. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI EA '19). Association for Computing Machinery, New York, NY, USA, 1–6. doi:10.1145/3290607.3313084
- [33] Alessandro Murgia, Daan Janssens, Serge Demeyer, and Bogdan Vasilescu. 2016. Among the machines: Human-bot interaction on social q&a websites. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*. 1272–1279. doi:10.1145/2851581.2892311
- [34] Shishir Nagaraja and Ryan Shah. 2019. Clicktok: Click fraud detection using traffic analysis. In *Proceedings of the 12th Conference on Security and Privacy in Wireless and Mobile Networks*. 105–116.
- [35] Anuar Nawawi and Ahmad Saiful Azlin Puteh Salin. 2018. Internal control and employees' occupational fraud on expenditure claims. *Journal of Financial Crime* 25, 3 (2018), 891–906. doi:10.1108/jfc-07-2017-0067
- [36] Hyunho Park, Hyoungju Kim, and Pan-Koo Kim. 2021. Development of Electronic Library Chatbot System Using SNS-Based Mobile Chatbot Service. In *The 9th International Conference on Smart Media and Applications* (Jeju, Republic of Korea) (SMA 2020). Association for Computing Machinery, New York, NY, USA, 364–366. doi:10.1145/3426020.3426134
- [37] Lara Schibelsky Godoy Piccolo, Pinelopi Troullinou, and Harith Alani. 2021. Chatbots to Support Children in Coping with Online Threats: Socio-Technical Requirements. In *Proceedings of the 2021 ACM Designing Interactive Systems Conference* (Virtual Event, USA) (DIS '21). Association for Computing Machinery, New York, NY, USA, 1504–1517. doi:10.1145/3461778.3462114
- [38] Natalia Pinzón, Vikram Koundinya, Ryan Galt, William Dowling, Marcela Boukloh, Namah C Taku-Forchu, Tracy Schohr, Leslie Roche, Samuel Ikendi, Mark H Cooper, et al. 2023. *AI-Powered Fraud and the Erosion of Online Survey Integrity: An Analysis of 31 Fraud Detection Strategies*. Technical Report. Center for Open Science.
- [39] Andreas Pitsillidis, Kirill Levchenko, Christian Kreibich, Chris Kanich, Geoffrey M Voelker, Vern Paxson, Nicholas Weaver, and Stefan Savage. 2010. Botnet Judo: Fighting Spam with Itself. In *NDSS*.
- [40] Manuel Portela and Carlos Granell-Canut. 2017. A new friend in our smartphone? Observing interactions with chatbots in the search of emotional engagement. In *Proceedings of the XVIII International Conference on Human Computer Interaction*. 1–7. doi:10.1145/3123818.3123826
- [41] Sanjeev Rao, Anil Kumar Verma, and Tarunpreet Bhatia. 2021. A review on social spam detection: Challenges, open issues, and future directions. *Expert Systems with Applications* 186 (2021), 115742.
- [42] Jacob Ratkiewicz, Michael Conover, Mark Meiss, Bruno Gonçalves, Snehal Patil, Alessandro Flammini, and Filippo Menczer. 2011. Truthy: mapping the spread of astroturf in microblog streams. In *Proceedings of the 20th international conference companion on World wide web*. 249–252. doi:10.1145/1963192.1963301
- [43] Mike Richards, Kevin Waugh, Mark Slaymaker, Marian Petre, John Woodthorpe, and Daniel Gooch. 2024. Bob or Bot: Exploring ChatGPT's Answers to University Computer Science Assessment. *ACM Transactions on Computing Education* 24, 1 (2024), 1–32.
- [44] Christopher Rodriguez and Daniel M Oppenheimer. 2024. Creating a Bot-tleneck for malicious AI: Psychological methods for bot detection. *Behavior Research Methods* (2024), 1–18.
- [45] Sayak Saha Roy, Poojitha Thota, Krishna Vamsi Naragam, and Shirin Nilizadeh. 2024. From Chatbots to Phishbots?: Phishing Scam Generation in Commercial Large Language Models. In *2024 IEEE Symposium on Security and Privacy (SP)*. IEEE Computer Society, 221–221.
- [46] Victoria L Rubin. 2017. Deception detection and rumor debunking for social media. In *The SAGE handbook of social media research methods*. Sage London, 342.
- [47] Johnny Saldaña. 2021. The coding manual for qualitative researchers. (2021).
- [48] Abigail See and Christopher D Manning. 2021. Understanding and predicting user dissatisfaction in a neural generative chatbot. In *Proceedings of the 22nd Annual Meeting of the Special Interest Group on Discourse and Dialogue*. 1–12. <https://aclanthology.org/2021.sigdal-1.1>
- [49] Joseph Seering, Juan Pablo Flores, Saiph Savage, and Jessica Hammer. 2018. The Social Roles of Bots: Evaluating Impact of Bots on Discussions in Online Communities. *Proc. ACM Hum.-Comput. Interact.* 2, CSCW, Article 157 (nov 2018), 29 pages. doi:10.1145/3274426
- [50] Joseph Seering, Michal Luria, Geoff Kaufman, and Jessica Hammer. 2019. Beyond dyadic interactions: Considering chatbots as community members. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [51] Benjamin T Sheehan. 2018. *Customer service chatbots: Anthropomorphism, adoption and word of mouth*. Ph. D. Dissertation. Queensland University of Technology. doi:10.5204/thesis.eprints.121188
- [52] Basia Spalek. 2017. *Crime victims: Theory, policy and practice*. Bloomsbury Publishing. doi:10.1177/026858090702200237
- [53] Gunnar Stevens, Alexander Boden, Fatemeh Alizadeh, Timo Jakobi, Michelle Walther, and Jana Krüger. 2022. Wie gehen verbrauch: innen mit onlinebetrug um?—Eine literaturübersicht. *Rüdiger, Bayerl (Hg.): Handbuch Cyberkriminalologie* (2022), 1–22. doi:10.1007/978-3-658-35450-3_42-1
- [54] Zachary K Stine and Nitin Agarwal. 2019. Characterizing the language-production dynamics of social media users. *Social Network Analysis and Mining* 9 (2019), 1–13.
- [55] Zachary K Stine, Tuja Khaund, and Nitin Agarwal. 2018. Measuring the information-foraging behaviors of social bots through word usage. In *2018 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*. IEEE, 570–571.
- [56] Richard Takacs and Ian McCulloh. 2019. Dormant bots in social media: Twitter and the 2018 US senate election. In *Proceedings of the 2019 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining*. 796–800. doi:10.1145/3341161.3343852
- [57] Hochschule Bonn-Rhein-Sieg Universität Siegen, open.INC GmbH. 2024. AntiScam: Abwehr von Kommunikationsbetrug zum Schutz der digitalen Identität. <https://www.forschung-it-sicherheit-kommunikationssysteme.de/projekte/antiscam>. Accessed: 2025-02-28.
- [58] Zijie J. Wang, Chinmay Kulkarni, Lauren Wilcox, Michael Terry, and Michael Madaio. 2024. Farsight: Fostering Responsible AI Awareness During AI Application Prototyping. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '24). Association for Computing Machinery, New York, NY, USA, Article 976, 40 pages. doi:10.1145/3613904.3642335
- [59] Max Weiss. 2019. Deepfake bot submissions to federal public comment websites cannot be distinguished from human submissions. *Technology Science* 2019121801 (2019).
- [60] Qian Xiao and Elisa Bertino. 2017. Detecting deceptive engagement in social media by temporal pattern analysis of user behaviors: a survey. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery* 7, 5 (2017), e1210.
- [61] Semih Yagcioglu, Mehmet Saygin Seyfioglu, Begum Citamak, Batuhan Bardak, Seren Guldamlasioglu, Azmi Yuksek, and Emin Islam Tatli. 2019. Detecting cybersecurity events from noisy short text. *arXiv preprint arXiv:1904.05054* (2019).
- [62] Haifeng Yu. 2011. Sybil defenses via social networks: a tutorial and survey. *ACM SIGACT News* 42, 3 (2011), 80–101. doi:10.1145/2034575.2034593
- [63] Ziyi Zhang, Shuofei Zhu, Jaron Mink, Aiping Xiong, Linhai Song, and Gang Wang. 2022. Beyond bot detection: combating fraudulent online survey takers. In *Proceedings of the ACM Web Conference 2022*. 699–709.
- [64] Mu Zhu, Ahmed H Anwar, Zelin Wan, Jin-Hee Cho, Charles A Kamhoua, and Munindar P Singh. 2021. A survey of defensive deception: Approaches using game theory and machine learning. *IEEE Communications Surveys & Tutorials* 23, 4 (2021), 2460–2493.